INCH-POUND

MIL-C-19803F
AMENDMENT 5
29 September 1995
SUPERSEDING
AMENDMENT 4
21 November 1986

MILITARY SPECIFICATION

CONVERTER, LIQUID OXYGEN, 10 LITER, GCU-24A/A

This amendment forms a part of MIL-C-19803F, dated 24 March 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

By this amendment, "GCU-24/A" is changed to "GCU-24A/A" wherever it appears in this specification.

By this amendment, "Overpressure safety" is changed to "Secondary overpressure safety" wherever it appears in this specification.

PAGE 2

* 2.1.1 SPECIFICATIONS, MILITARY: Delete the following:

"MIL-C-81302 Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-T-81533 Trichloroethane 1, 1, 1 (Methyl Chloroform) Inhibited, Vapor Degreasing"

2.1.1 STANDARDS, MILITARY: Delete "MIL-STD-781 Reliability Tests-Exponential Distribution" and substitute "MIL-STD-781 Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution."

PAGE 3

* 2.1.1 STANDARDS, MILITARY: Add "MIL-STD-1359 Cleaning Methods and Procedures for Breathing Oxygen Equipment."

PAGE 4

3.6 List of major components: Add "Mounting base" to the end of list.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.

PAGE 5

3.6.4, lines 6 and 7: Delete third sentence and substitute "The sensing element shall be an integral part of the container and shall be permanently installed."

3.6.10 Delete and substitute the following:

"3.6.10 Secondary overpressure safety device. A secondary overpressure safety device, incorporating a frangible safety burst disc, shall be embodied in the plumbing of the converter. The location of the safety device shall be such that it will ensure rapid dumping of the head pressure from the converter in the event the pressure relief valve fails to open. The safety device shall contain a visible means of ascertaining the condition of the frangible safety burst disc without disassembly."

PAGE 6

Add the following new paragraph:

"3.6.13 Mounting base. The mounting base shall conform to the configurations and dimensional requirements of Figure 4."

PAGE 7

3.7.10.1 Delete and substitute the following:

"3.7.10.1 Secondary overpressure safety (standard temperature). The secondary overpressure safety device, when tested as specified in 4.8.11.1, shall not leak at a pressure below 175 psig (1206.58 kPa) and the frangible safety burst disc (see 3.6.10) shall rupture within a pressure range of 200 psig (1378.95 kPa) minimum and 250 psig (1723.68 kPa) maximum. The safety device shall provide a minimum flow of 150 lpm with the safety disc in the ruptured condition."

3.7.10.2 Delete and substitute the following:

"3.7.10.2 Secondary overpressure safety (low temperature). The secondary overpressure safety device, when tested as specified in 4.8.11.2, shall not leak at a pressure below 175 psig (1206.58 kPa) and the frangible safety burst disc (see 3.6.10) shall rupture within a pressure range of 200 psig (1378.95 kPa) minimum and 315 psig (2171.84 kPa) maximum. The safety device shall provide a minimum flow of 150 lpm with the safety disc in the ruptured condition."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.

PAGE 7 (Continued)

3.7.10.3 Delete and substitute the following:

"3.7.10.3 Secondary overpressure safety (high temperature). The secondary overpressure safety device, when tested as specified in 4.8.11.3, shall not leak at a pressure below 175 psig (1206.58 kPa) and the frangible safety burst disc (see 3.6.10) shall rupture within a pressure range of 175 psig (1206.58 kPa) minimum and 250 psig (1723.68 kPa) maximum. The safety device shall provide a minimum flow of 150 lpm with the safety disc in the ruptured condition."

PAGE 9

* 3.11.1 Delete and substitute the following:

"3.11.1 Degreasing. Prior to assembling the converter, all internal surfaces of the converter shall be degreased and cleaned in accordance with MIL-STD-1359. After assembly, the absence of cleaning compounds shall be verified in accordance with MIL-STD-1359."

PAGE 10

* 3.13.2 Delete the following from the end of the paragraph:

"If a removable probe is employed in the container, another note shall be located in the container to designate the applicable probe number.

WARNING

USE PROBE NUMBER ONLY"

PAGE 11

4.3.1 Delete items a through e and substitute the following:

"a. Two each complete converter assemblies.
b. One each inner vessel complete with all lines and fittings.
c. One each combination fill-buildup-vent valve.
d. One each relief valve.
e. One each pressure control valve.
f. Three each secondary overpressure safety devices.
g. A Statement of Certification (with supporting test data) from the manufacturer that the converters conform to the reliability requirements cited in paragraphs 3.7.16 and 4.8.17."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.

PAGE 14

4.6.1: Add the following to the list of tests and examinations:

"Dimensions
Relief valve test"

PAGE 19

4.8.11, line 4: Delete "opening" and substitute "rupturing of the frangible safety burst disk".

PAGE 26

Add the following new paragraph:

"6.8 Units conversion. Conversion of units in this document is in accordance with ASTM E380-82 Metric Practice. The following conversion factors are applicable to this specification.

Atmosphere x 101.3 = kilopascals (kPa)
Pounds per square inch (psi) x 6.894 = kPa
Inches x 25.4 = millimeters (mm)
Feet x 0.3048 = meters (m)
Liters per minute (lpm) x 0.001 = cubic meters per minute (cu m/min)
Pounds x 0.4535 = kilogram (kg)
Inches of Hg x 3.376 = kPa
[Degrees Fahrenheit (°F) - 32] / 1.8 = Degrees Celcius (°C)"

PAGE 29

Figure 3: Delete and substitute new figure 3.

PAGE 30

Figure 4: Delete and substitute new figure 4.

PAGE 32

Figure 6, Double Amplitude Inch (vertical scale): Delete ".0012" and substitute ".0001".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.

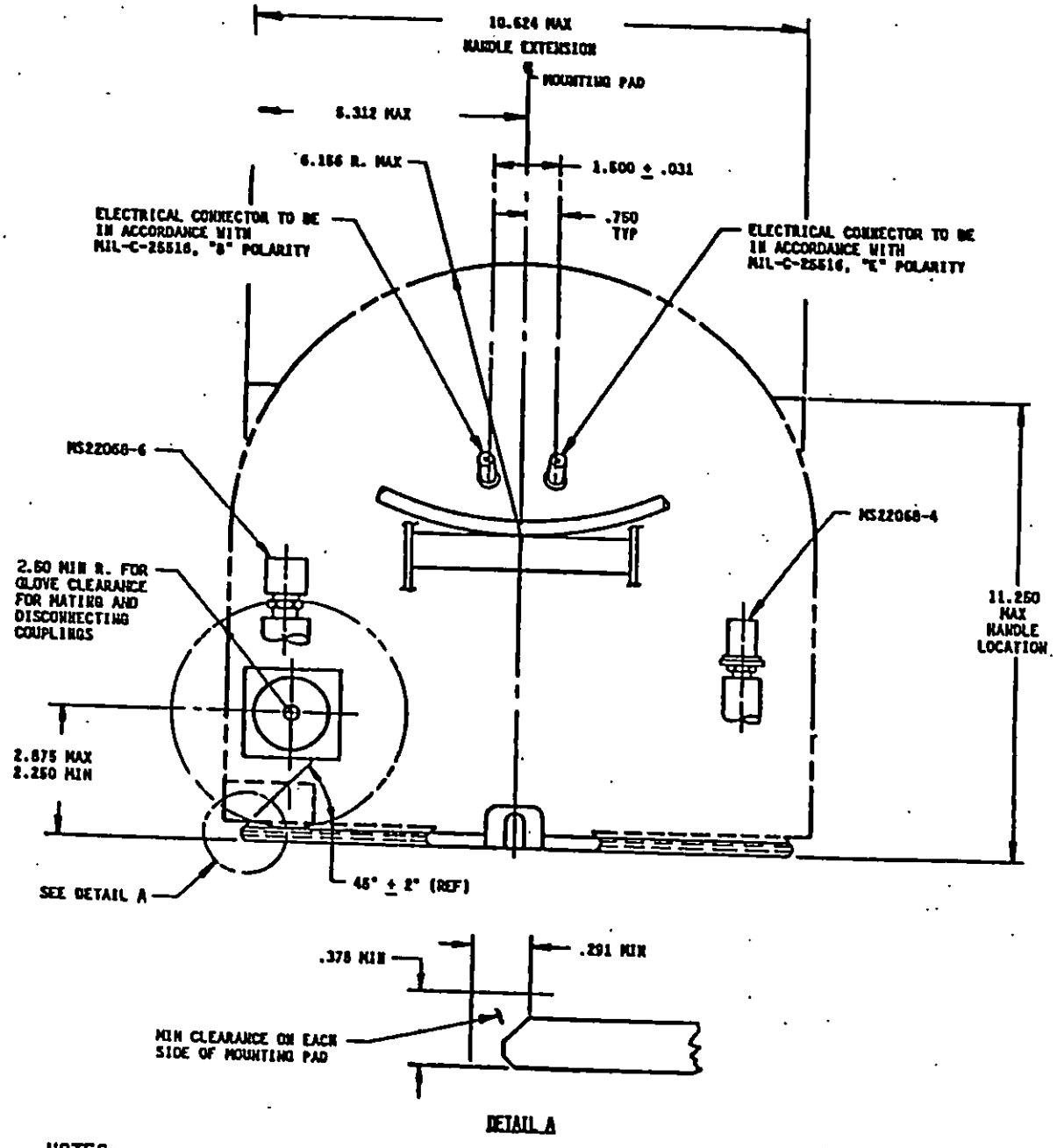
Custodians:
Army - AS
Air Force - 99

Review Activity:
Air Force - 71

Preparing Activity:
Navy - AS

(Project 1660-0691)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.



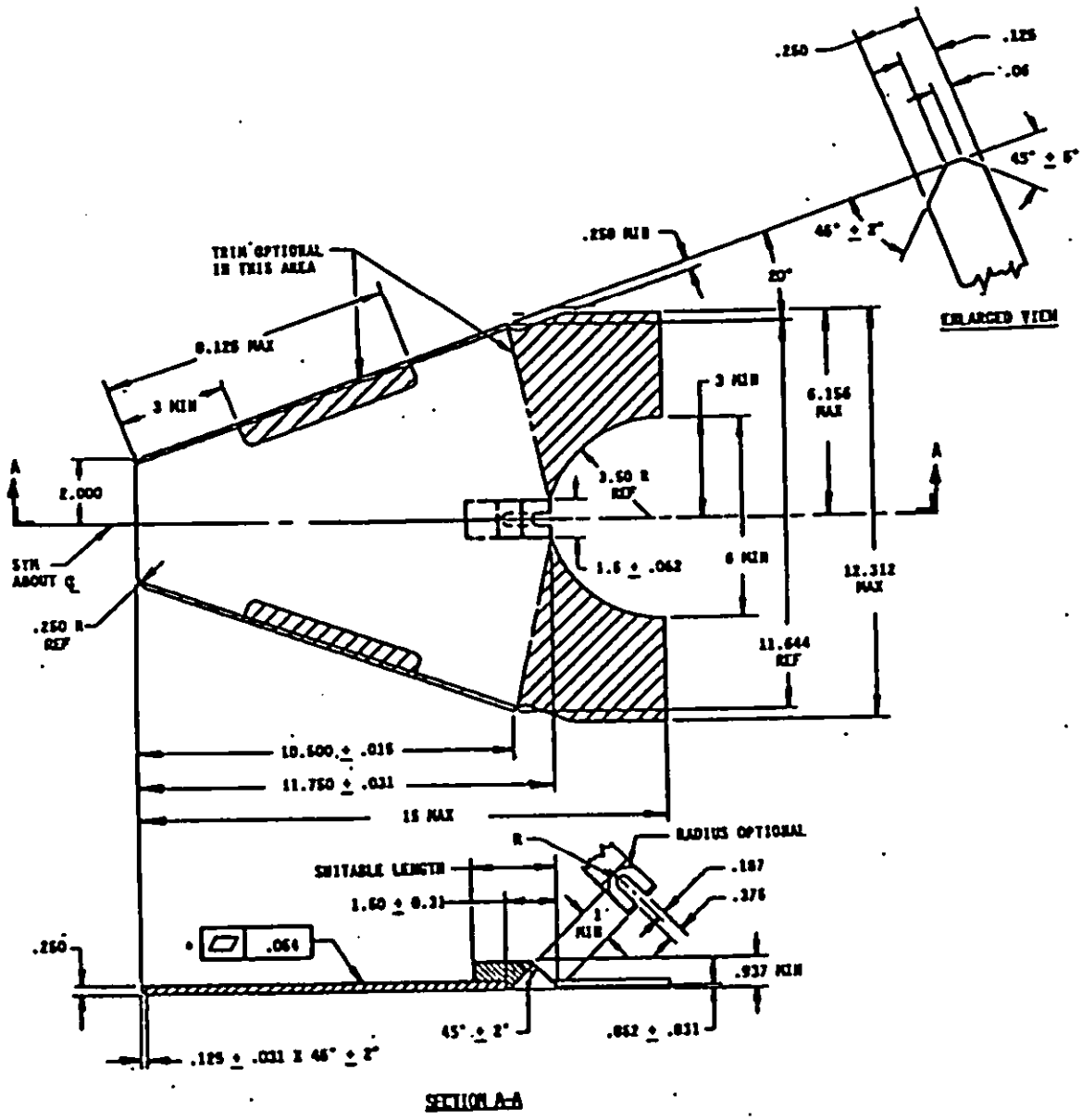
DETAIL A

NOTES:

1. All components shall be within the specified maximum envelope.
2. Unless otherwise specified, dimensions in inches.

FIGURE 3. Front View

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.



SECTION A-A

NOTES:

1. This mounting base is designed to mate with the mounting bracket shown on MS90341.
2. Unless otherwise specified, dimensions in inches. Tolerances: Decimals ± .010, Angles ± 1/2°.

FIGURE 4. Mounting base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:41Z
Check the source to verify that this is the current version before use.